# 张东宽 简历Curriculum Vitae — Dongkuan Zhang

## 基本信息Personal Information

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| **姓名Name** | 张东宽Dongkuan Zhang |
| **国籍Nationality** | 中国Chinese |
| **出生日期Date of Birth** | 02.05.1995 |
| **所在地Location** | 辽宁大连Dalian, Liaoning, China |
| **电话Phone** | +86 15510689505 |
| **邮箱Email** | dkz@mail.dlut.edu.cn |
| **语言Languages** | 中文（母语）、英语（流利）Mandarin (Native), English (Fluent) |

## 教育经历Education

**2023 – 今Now** **大连理工大学Dalian University of Technology**

环境科学与工程，博士（预计 2027 年毕业），导师：姬国钊 教授Ph.D. in Environmental Science and Engineering (expected 2027). Supervisor: Prof. Guozhao Ji

**2020 – 2023** **中央民族大学Minzu University of China**

环境科学与工程，硕士M.S. in Environmental Science and Engineering

**2013 – 2017** **辽宁石油化工大学Liaoning Petrochemical University**

石油工程，学士B.S. in Petroleum Engineering

## 工作经历Work Experience

**2017 – 2019** **浙江石油化工有限公司Zhejiang Petrochemical Co., Ltd.**

柴油加氢裂化装置操作员。负责装置监控、工艺调整、采样分析、故障处置、安全管理与记录维护。Diesel hydrocracking unit operator — equipment monitoring, process adjustment, sample collection and analysis, troubleshooting, safety management and record keeping.

## 联合培养 / 访学经历Joint Training & Visiting

**2025 – 2026** **东京科学大学（原东京工业大学）Cross 实验室Cross Laboratory, Institute of Science Tokyo**

联合培养博士生，导师：Jeffrey S. Cross 教授。资助：国家公派留学（CSC）。Joint PhD student supervised by Prof. Jeffrey S. Cross. Funded by the China Scholarship Council (CSC).

**2026.08 – 2027.02** **卢森堡大学University of Luxembourg**

联合培养博士生（6 个月）。资助：大连理工大学"连理全球"留学专项。Joint PhD research visit (6 months). Funded by the DUT Global Vision Project.

## 研究方向Research Interests

- **面向废弃物焚烧技术的智能多相流建模**：将基于物理机理的 CFD 数值模拟与 AI 驱动的流场预测相结合。**Intelligent multiphase flow modeling for waste incineration**: combining physics-based CFD with AI-driven flow field prediction.
- 多相流、燃烧与传热过程的欧拉-拉格朗日数值模拟（实验室规模至工业规模）。Eulerian-Lagrangian modeling of multiphase flow, combustion and heat transfer (lab to industrial scale).
- 机器学习与神经网络代理模型（输入凸神经网络等）用于流场快速预测与非牛顿流体建模。Machine-learning and neural-network surrogates (e.g. input-convex neural networks) for fast flow prediction and non-Newtonian fluid modeling.
- 焚烧炉、循环流化床、双流体喷嘴等装备的流场设计、防堵与运行优化。Flow field design, anti-clogging and operational optimization of incinerators, circulating fluidized beds and twin-fluid nozzles.

## 学术成果概览Academic Output Snapshot

总被引Total citations **60** · h-index **4** · i10-index **1** (更新于updated 2026-09-01)

- 期刊论文Journal articles: **11** 篇papers（其中第一作者 / 共一incl. first/co-first author 5 篇papers）
- 专利Patents: **8** 项patents（含发明专利申请与实用新型including invention applications and utility models）
- 软件著作权Software copyrights: **2** 项items
- 团体标准Group standards: **1** 项（参编）(participating drafter)
- 科技成果转化Technology transfer: **1** 项，成果转让金额 40 万元patent transfer, RMB 400,000

## 期刊论文Journal Articles (11)

**2026**

**Clogging mechanism and early-warning criterion for shear-thinning distillation residues in twin-fluid spray nozzles based on gas-liquid multiphase flowClogging mechanism and early-warning criterion for shear-thinning distillation residues in twin-fluid spray nozzles based on gas-liquid multiphase flow**

D Zhang, T Ren, Y Liu, M Li, J Yang, J Dai, J Li, JS Cross, G JiD Zhang, T Ren, Y Liu, M Li, J Yang, J Dai, J Li, JS Cross, G Ji

*Chemical Engineering Research and DesignChemical Engineering Research and Design* · link

**2026**

**Numerical Investigation of Coal and Rice Husk Co-Combustion in an Industrial-Scale Circulating Fluidized Bed: Hydrodynamics, Temperature, and Pollutant EmissionsNumerical Investigation of Coal and Rice Husk Co-Combustion in an Industrial-Scale Circulating Fluidized Bed: Hydrodynamics, Temperature, and Pollutant Emissions**

L Liu, J Sun, YS Zhang, T Anjum, D Zhang, J Yang, J Dong, S Cheng, G JiL Liu, J Sun, YS Zhang, T Anjum, D Zhang, J Yang, J Dong, S Cheng, G Ji

*Processes 14 (13), 2189Processes 14 (13), 2189* · link

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| **2026** | **Quantifying the Influence of Supercritical Drying on the Pore Size Distribution of Cobalt-Doped Silica MembraneQuantifying the Influence of Supercritical Drying on the Pore Size Distribution of Cobalt-Doped Silica Membrane**<br><br>T Anjum, D Zhang, G Olguin, B Ladewig, DK Wang, X Zhang, AL Khan, ...T Anjum, D Zhang, G Olguin, B Ladewig, DK Wang, X Zhang, AL Khan, ...<br><br>*Industrial & Engineering Chemistry ResearchIndustrial & Engineering Chemistry Research* · link |
| **2026** | **Impact of freeze drying on pore size distribution of amorphous silica membranes derived from gas permeation activation energiesImpact of freeze drying on pore size distribution of amorphous silica membranes derived from gas permeation activation energies**<br><br>T Anjum, TH Qin, G Olguin, Y Wang, D Zhang, X Kou, DK Wang, X Zhang, ...T Anjum, TH Qin, G Olguin, Y Wang, D Zhang, X Kou, DK Wang, X Zhang, ...<br><br>*Separation and Purification Technology, 137736Separation and Purification Technology, 137736* · link · 2 citations |
| **2025** | **The Eulerian–Lagrangian Model for Simulating the Moisture Content Effect on the Characteristics of MSW Combustion in a 50 T/D Grate IncineratorThe Eulerian–Lagrangian Model for Simulating the Moisture Content Effect on the Characteristics of MSW Combustion in a 50 T/D Grate Incinerator**<br><br>J Dai, Y Du, Y Xie, D Zhang, L Liu, Y Gui, G JiJ Dai, Y Du, Y Xie, D Zhang, L Liu, Y Gui, G Ji<br><br>*Processes 13 (12), 3928Processes 13 (12), 3928* · link |
| **2025** | **Eulerian-Largrangian model for simulation of flow behavior and heat transfer of lab scale dual circulating fluidized bedsEulerian-Largrangian model for simulation of flow behavior and heat transfer of lab scale dual circulating fluidized beds**<br><br>J Yang, D Zhang, L Liu, Y Xie, Y Du, A Li, C Qin, G JiJ Yang, D Zhang, L Liu, Y Xie, Y Du, A Li, C Qin, G Ji<br><br>*Applied Thermal Engineering, 127937Applied Thermal Engineering, 127937* · link · 5 citations |
| **2025** | **Simulation of multiphase flow with thermochemical reactions: A review of computational fluid dynamics (CFD) theory to AI integrationSimulation of multiphase flow with thermochemical reactions: A review of computational fluid dynamics (CFD) theory to AI integration**<br><br>D Zhang, T Anjum, Z Chu, JS Cross, G JiD Zhang, T Anjum, Z Chu, JS Cross, G Ji<br><br>*Renewable and Sustainable Energy Reviews 221, 115895Renewable and Sustainable Energy Reviews 221, 115895* · link · 39 citations |
| **2023** | **深海多金属结核采集过程对沉积物扰动试验研究Experimental study on sediment disturbance during deep-sea polymetallic nodule collection**<br><br>张东宽， 刘美麟， 夏建新Dongkuan Zhang, Meilin Liu, Jianxin Xia<br><br>*矿冶工程 43 (3)Mining and Metallurgical Engineering 43 (3)* · link · 4 citations |

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| **2022** | **Flow field parameter matching of deep-sea polymetallic nodule fluidization acquisition headFlow field parameter matching of deep-sea polymetallic nodule fluidization acquisition head**<br><br>DZLGHCJ Xia.DZLGHCJ Xia.<br><br>*Journal of Marine Technology 41 (4), 53-60Journal of Marine Technology 41 (4), 53-60* · link |
| **2021** | **Comparison and analysis of deep-sea polymetallic nodule collection methods and structural parametersComparison and analysis of deep-sea polymetallic nodule collection methods and structural parameters**<br><br>L Guan, D Zhang, Y Xia, J XiaL Guan, D Zhang, Y Xia, J Xia<br><br>*J. Ocean Technol 40 (5), 62-70J. Ocean Technol 40 (5), 62-70* · link · 7 citations |
| **2021** | **深海多金属结核采集方式及其结构参数比较分析Comparative analysis of collection methods and structural parameters for deep-sea polymetallic nodules**<br><br>官雷， 张东宽， 夏玉超， 夏建新Lei Guan, Dongkuan Zhang, Yuchao Xia, Jianxin Xia<br><br>*海洋技术学报 40 (5), 62Journal of Ocean Technology 40 (5), 62* · link · 3 citations |

查看完整出版列表（含引用数）See the full publication list with citation counts

## 专利Patents (10)

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| **2026** | **一种外混式双流体雾化喷嘴An external-mixing dual-fluid atomizing spray nozzle**<br><br>姬国钊, 张东宽Guozhao Ji, Dongkuan Zhang<br><br>*中国发明专利，申请号 CN121847357AChina Invention Patent Application No. CN121847357A* |
| **2026** | **一种自清洁双喷嘴系统A self-cleaning dual-spray nozzle system**<br><br>姬国钊, 张东宽, 杨延军, 戴家诚, 田红, 任桐Guozhao Ji, Dongkuan Zhang, Yanjun Yang, Jiacheng Dai, Hong Tian, Tong Ren<br><br>*中国发明专利，申请号 CN121972345AChina Invention Patent Application No. CN121972345A* |
| **2025** | **一种垃圾焚烧炉内自降飞灰的装置A device for self-settling fly ash in a waste incinerator**<br><br>姬国钊, 张东宽Guozhao Ji, Dongkuan Zhang<br><br>*中国实用新型专利，ZL 2024 2 2052228.7（授权公告号 CN 223137888 U）China Utility Model Patent No. ZL 2024 2 2052228.7 (Granted CN 223137888 U)* |
| **2024** | **A fluidized bed reactor based on Tesla Valve**<br><br>GJD Zhang<br><br>*CN Patent CN202311155946.0* · Scholar |

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| **2024** | **Device, method and Application of self-lowering fly Ash in waste incinerator**<br><br>GJD Zhang<br><br>*CN Patent CN202411163901.2* · Scholar |
| **2024** | **一种垃圾焚烧炉内自降飞灰的装置、方法及应用A device, method and application for self-settling fly ash in a waste incinerator**<br><br>姬国钊, 张东宽Guozhao Ji, Dongkuan Zhang<br><br>*中国发明专利申请公开号 CN118912512AChina Invention Patent Application Publication No. CN118912512A* |
| **2023** | **一种垃圾焚烧炉内自降飞灰的装置、方法及应用A device, method and application for self-settling fly ash in a waste incinerator**<br><br>姬国钊, 张东宽Guozhao Ji, Dongkuan Zhang<br><br>*中国发明专利申请公开号 CN117308105AChina Invention Patent Application Publication No. CN117308105A* |
| **2022** | **A design method for flow field of polymetallic nodules collector**<br><br>JXLGFLHCDWD Zhang<br><br>*CN Patent CN202111566075.2* · Scholar |
| **2022** | **A flow field design method for polymetallic nodules collector**<br><br>JXDZFLHCDWL Guan<br><br>*CN Patent CN202111562069. X* · Scholar |
| **2016** | **A kind of probe for petroleum exploration**<br><br>CWBZSMTWZWYSCLSPWDZJLCADWSYH Sun<br><br>*CN Patent CN201620273762.3* · Scholar |

## 软件著作权Software Copyrights

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| **2025-09** | **双喷枪自清洁控制系统 DSGSCS V1.0Dual Spray Gun Self-Cleaning Control System (DSGSCS) V1.0**<br><br>*China Computer Software Copyright Registration* |
| **2025-07** | **喷嘴（管道）堵塞风险监测软件 NRMS V1.0Nozzle (Pipe) Blockage Risk Monitoring Software (NRMS) V1.0**<br><br>*China Computer Software Copyright Registration* |

## 标准Standards

**2025** **《基于高温热解气化的废弃物制氢技术要求》Technical Requirements for Hydrogen Production from Waste Based on High-Temperature Pyrolysis and Gasification**

张东宽（参编）等Dongkuan Zhang (participating drafter), et al.

*Group/Industry Standard (participating author)*

## 科研项目与科技成果转化Research Projects & Technology Transfer

**2023 – 2024** **垃圾焚烧炉自降飞灰技术研发（科技成果转化）Self-settling fly ash technology for waste incinerators (technology transfer)**

作为成果完成人之一，将自主研发的专利成果及相关研究成果转让给企业，转让金额 40 万元；个人获学校现金奖励 2 万元。As one of the completing researchers, transferred self-developed patent achievements and related research results to industry (RMB 400,000). Personally awarded RMB 20,000 cash incentive by the university.

## 荣誉与获奖Honors & Awards

**2026.04** **大连理工大学"连理全球"留学专项资助**

DUT Global Vision Project Funding (University of Luxembourg, 6-month joint PhD visit).

**2025.12** **教育部 2025 春晖海外青年优秀成果（创新创业类）**

2025 Chunhui Overseas Young Excellent Achievement Collection (MOE of China). Project: "Crisis-Flow Control: Flow-field-based Hazardous Waste Incineration & Disposal Technology and Equipment."

**2025.09** **博士研究生国家奖学金**

National Scholarship for Doctoral Students (Ministry of Education of China).

**2025.11** **第 12 届中日韩清洁能源与可持续环境研讨会（日本山梨）报告证书**

12th Japan-China-Korea Joint Symposium on Clean Energy and Sustainable Environment, Yamanakako, Japan — Presentation Certificate.

**2025.01** **团体标准 T/CITS 242-2025《基于高温热解气化的垃圾制氢技术要求》起草人证书**

Drafter Certificate, Group Standard T/CITS 242-2025, China Inspection and Testing Society.

**2024.12** **大连理工大学 2024 年度"环境工程之星"**

"Environmental Engineering Star" (Dalian University of Technology).

**2024.11** **2024 韩中日能源环境联合研讨会（韩国延世大学）优秀报告奖**

Award for Excellent Presentation, 2024 Korea-China-Japan Joint Symposium on Energy and Environment (Yonsei University).

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| **2024.11** | **国家公派留学资格（CSC 公派证书）**<br><br>China Scholarship Council (CSC) Government-Sponsored Studentship (CSC No. 202406060182). |
| **2024** | **科技成果转化现金奖励（2 万元）**<br><br>Cash incentive for technology transfer (RMB 20,000), Dalian University of Technology. |

## 技术服务与学术兼职Service & Professional Activities

- 团体标准 T/CITS 242-2025《基于高温热解气化的垃圾制氢技术要求》起草人（中国检验检测学会）。Drafter of group standard T/CITS 242-2025, China Inspection and Testing Society.
- 同行评审：Chemical Engineering Research and Design、Separation and Purification Technology、Applied Thermal Engineering、Journal of Cleaner Production 等期刊审稿人。Peer reviewer for Chemical Engineering Research and Design, Separation and Purification Technology, Applied Thermal Engineering, Journal of Cleaner Production, and related journals.

## 技术技能Technical Skills

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| **CFD 模拟CFD** | ANSYS Fluent、Star-CCM+、COMSOL Multiphysics；多相流、燃烧、传热建模。ANSYS Fluent, Star-CCM+, COMSOL Multiphysics; multiphase flow, combustion and heat transfer modeling. |
| **编程Programming** | Python（含 PyTorch、NumPy、SciPy）、MATLAB；数据分析与可视化。Python (PyTorch, NumPy, SciPy), MATLAB; data analysis and visualization. |
| **实验Experiment** | 流场测试、传热实验、采样与表征。Flow-field testing, heat-transfer experiments, characterization. |